

Akash Karthik

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EDUCATION

Bachelor of Science, Computer Science

Expected May 2027

San Jose State University, San Jose, CA

Relevant Coursework: Data Structures & Algorithms, Linear Algebra, Discrete Math, Computer Systems, Intro to Circuits

TECHNICAL SKILLS

Flight & Autonomy: PX4 flight stack, flight-control firmware, visual-inertial odometry, IMU sensor fusion, state estimation, flight testing, telemetry log analysis

Controls: PID and fixed-point PI, gain tuning, anti-windup, saturation, slew-rate limiting, deterministic real-time execution

Simulation & Test: Physics-based vehicle simulation, software-in-the-loop replay, regression suites, parameter sweeps, headless batch runs, pytest

Languages & Tools: Python, C/C++, NumPy, pandas, Matplotlib, OpenCV, CAN/DBC, GDB, BusMaster, MoTeC i2 Pro, Git

EXPERIENCE

Software Engineer Intern

Feb 2026 - Present

Argus Defense | San Jose, CA

- Developed and validated embedded flight-control firmware in C on PX4, iterating control gains against logged flight telemetry
- Integrated and tuned visual-inertial odometry for onboard state estimation and evaluated performance against logged telemetry
- Tuned PX4 PID flight-control loops to reduce vertical and horizontal drift during autonomous position hold, iterating through repeatable flight tests and log review

Software Lead

Jun 2026 - Present

Spartan Racing, Formula SAE | San Jose, CA

- Lead validation strategy for vehicle controls, telemetry, diagnostics, simulation, and on-vehicle test workflows
- Define testable software requirements and coordinate reviews across controls, dashboard, simulation, and software-in-the-loop projects

Software Controls Engineer

Jul 2025 - Jun 2026

Spartan Racing, Formula SAE | San Jose, CA

- Implemented a torque-to-power controller with fixed-point PI control, saturation, anti-windup, and slew limits for deterministic real-time execution
- Built Python telemetry parsers and a software-in-the-loop replay harness to validate controller changes against recorded logs before hardware testing
- Planned and executed on-vehicle tests, measured response and latency, and turned results into parameter changes and regression checks

Software Intern

Aug 2024 - Jun 2025

Spartan Racing, Formula SAE | San Jose, CA

- Extracted a desktop simulator and regression suite to reproduce field scenarios, run controller sweeps, and prevent regressions
- **Results:** 1st in endurance at MIS EV 2025

PROJECT EXPERIENCE

SR-Wireless-CAN — Wireless Firmware Flashing & Telemetry

2026

Spartan Racing, Formula SAE | San Jose, CA | Backend lead

Python, python-can, bus trace analysis, FastAPI, WebSockets, SQLite, Docker

- Proposed and implemented wireless firmware flashing for the vehicle control unit, decoding the undocumented flashing sequence from bus trace captures and reimplementing it in Python
- Led backend development for the telemetry platform, streaming live signals to a browser dashboard with recorded-trace replay for offline test review

Vehicle Dynamics Simulation

Jul 2025

Spartan Racing, Formula SAE | San Jose, CA

Python, Pygame, Matplotlib, vehicle dynamics, PID control, CSV logging

- Built a physics-based simulator for torque, drag, acceleration, and steering to evaluate control changes before committing them to hardware
- Added headless batch runs, seeded scenarios, CSV summaries, and plots for repeatable offline comparison between builds